

CEIR11 Basic Civil Engineering
Module 4 : Water Supply

COURSE MENTOR

Dr. Aneesh Mathew

Assistant Professor

Department of Civil Engineering

National Institute of Technology, Tiruchirappalli

DELIVERED BY Ajisha S

Research Scholar (403122001)

Department of Civil Engineering

National Institute of Technology, Tiruchirappalli

Course Content

- Sources of Water – Characteristics of Water – Water Supply – Quality of Water- Wastewater Treatment

Water

- Elixir of life
- Is a vital natural resource that forms the basis of all life.
- **97%** of the water present on earth surface is salt water.
- Fresh water is renewable resources like soil and air.
- The world is supplied by clean and fresh water and it is decreasing. Water is one of our most critical resources, but around the world it is under threat.
- **Water demand** already exceeds supply in many parts of the world and as the **world population** continues to rise, so too does the water demand.







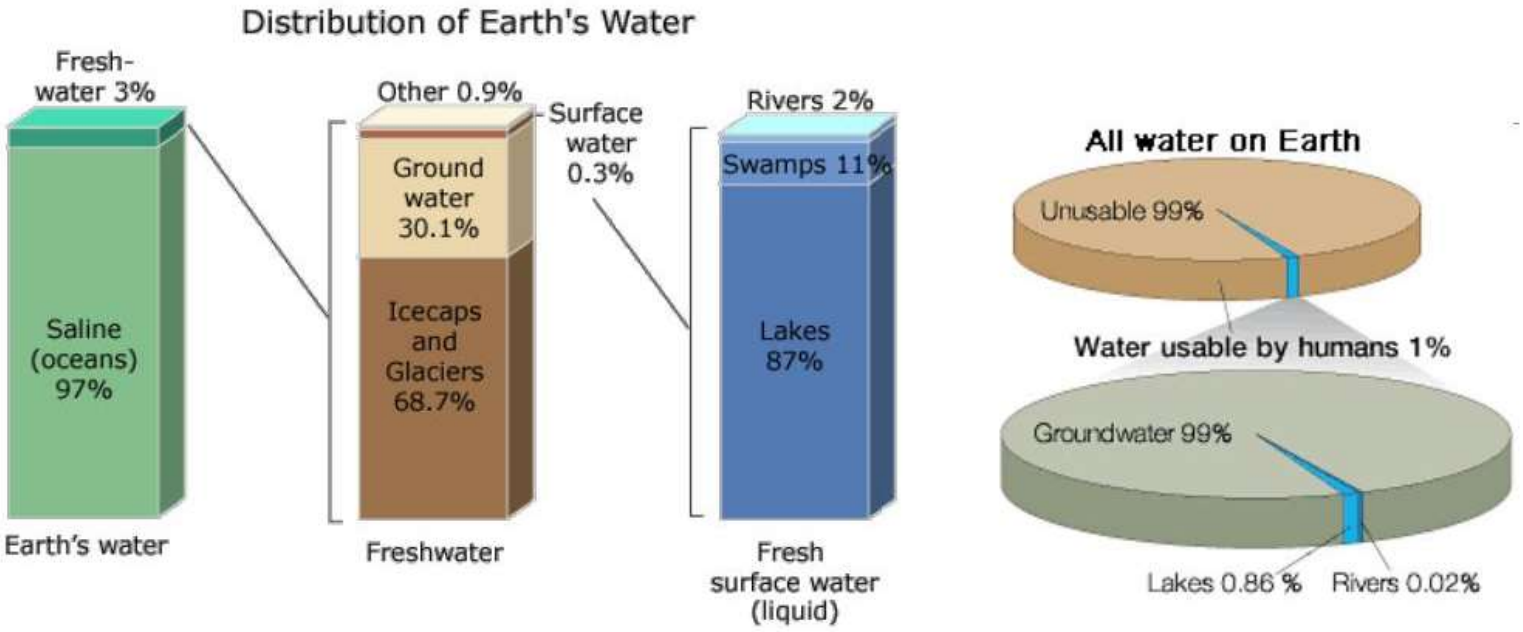


Water resources

- Water resources are sources of water that are useful or potentially useful to humans.
- It is important because it is needed for life to exist.
- Uses of water include agricultural, industrial, household, recreational and environmental activities.
- Virtually all of these human uses require fresh water.
- Only 2.5% of water on the Earth is fresh water, and over two thirds of this is frozen in glaciers and polar ice caps.

Distribution of water on Earth

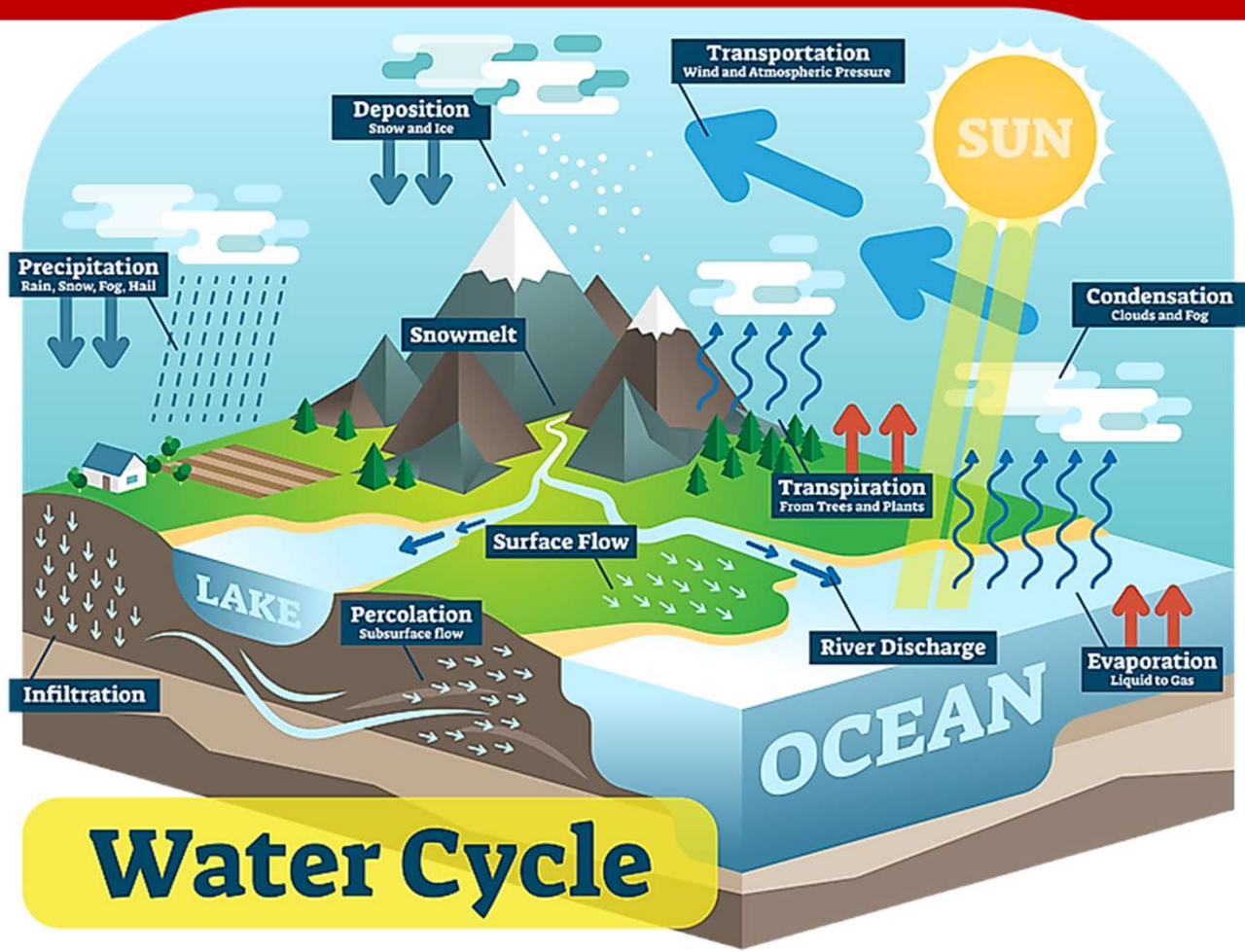
- Total water available = 1300 million cu.km



Source: USGS

Sources of water

- Water resources are divisible into two distinct categories : the surface-water resources and the ground-water resources.
- Each of these categories is a part of the earth's water circulatory system, called the hydrologic cycle, and is derived from precipitation, which is rainfall plus snow.



Surface water resources

- Surface water is water in a river, lake or fresh water wetland.
- Surface water is naturally replenished by precipitation and naturally lost through discharge to the oceans, evaporation, and sub-surface seepage.







Ground Water

- Sub-surface water, or groundwater, is fresh water located in the pore space of soil and rocks. It is also water that is flowing within aquifers below the water table.
- Springs, well etc.





Desalination

- Desalination is an artificial process by which saline water (generally sea water) is converted to fresh water



Frozen Water

- Several schemes have been proposed to make use of icebergs as a water source, however to date this has only been done for novelty purposes.
- Glacier runoff is considered to be surface water.

Uses of water

- Household and personal use - It is estimated that **15% of worldwide water use is for household purposes**. These include drinking water, bathing, cooking, sanitation, and gardening
- Washing dishes
- Watering the plants
- Putting out the fire
- Give water for the cows
- Watering the grass
- Washing the car

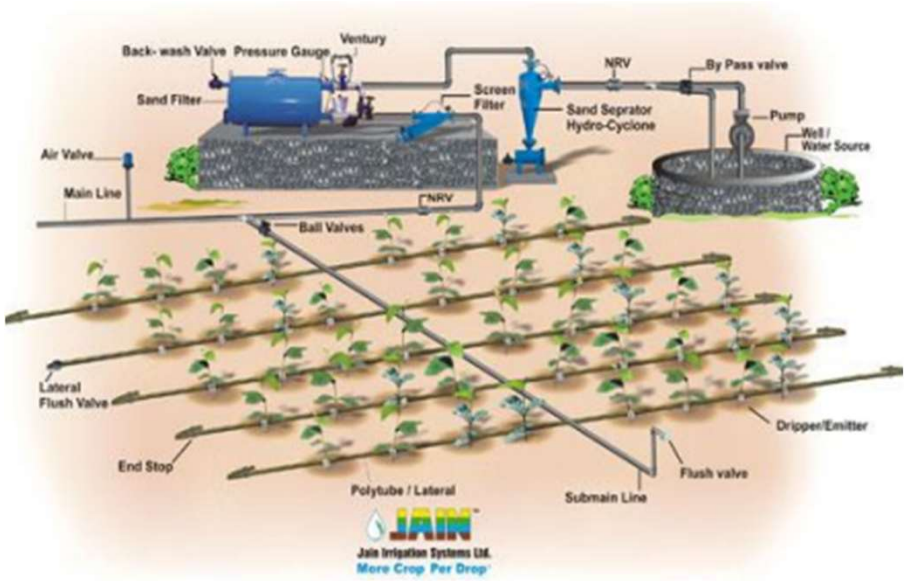




Uses of water

- Agricultural: It is estimated that **69% of worldwide water use is for irrigation**, with 15-35% of irrigation withdrawals being unsustainable. Aquaculture is a small but growing agricultural use of water.





Uses of water

- Industrial: It is estimated that **15% of worldwide water use is industrial**. The distribution of industrial water usage that is varies widely, but as a whole is lower than agricultural use.



Uses of water

- Recreational activities: use is usually a very small but growing percentage of total water use. Recreational water use is mostly tied to reservoirs
- Go to the pool.
- Go to the beach.
- Walk in the pool for recovering health.
- Skiing in the mountains.
- Fish in the lakes.
- Play with a ball in the river.
- Surfeit in the ocean.





- **Environmental:** Explicit environmental water use is also a very small but growing percentage of total water use



Per Day Water Consumption

Water footprint

Activity	Rate of consumption per day	Water used per day (approx)
Drinking	9-10 glasses	1.5 litres
Handwashing	4 times	8 litres
Flushing	2 times	20 litres
Bathing	2 times	240 litres
Cooking	3 times	6 litres
Washing vegetables, fruits and utensils	2-3 times	36 litres
Brushing	2 times	2 litres
Washing clothes	Once	80 litres
All of the above activities		**393.5 litres**

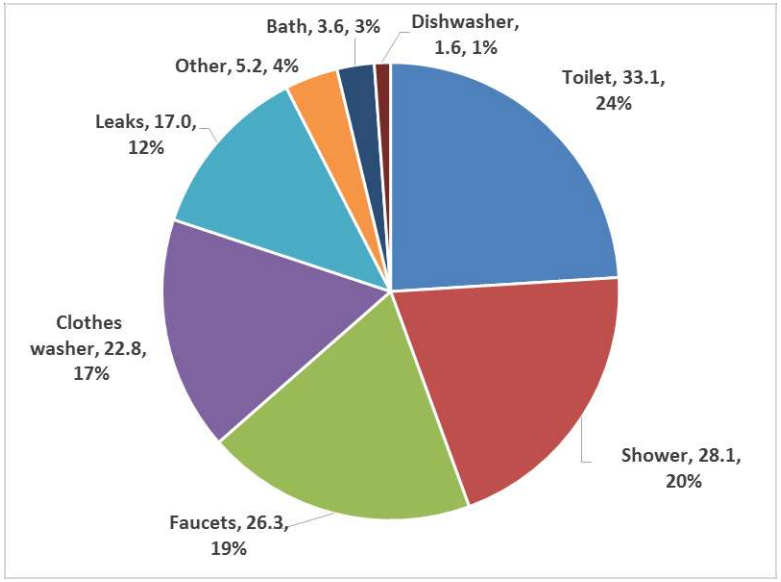
The details of the domestic consumption are

a) Drinking ----- 5 litres
b) Cooking ----- 5 litres
c) Bathing ----- 55 litres
d) Clothes washing -----20 litres
e) Utensils washing -----10 litres
f) House washing ----- 10 litres

---

135 litres/day/capita

Min Usage of water per day



- https://watercalculator.org/

- Blue, Green and Grey Water

Water Conservation Tips

- Run your dishwasher for only full loads.
- Turn off the tap when you brush your teeth.
- Check your faucets and toilets for leaks.
- Take shorter showers.
- Water your lawn in the morning.
- Use a cover on your pool.
- Wash your car with a nozzle on the hose.



We are fortunate enough to live in a place where clean water flows freely on demand, it can be easy to take it for granted. From your daily dishwashing routine to your daily shower, every drop of this natural resource is precious. Practicing conservation is not only good for the environment, but can save you money in energy bills.



Characteristics of Water

- **Precipitation** in the form of **rain** is the main source of water
- It may contain **trace amounts** of mineral matter, gases, and other substances as it forms and falls through the earth's atmosphere.
- The precipitation, however, has virtually **no bacterial content**.
- Once precipitation reaches the earth's surface, mineral and organic substances, micro organisms, and other forms of pollution (contamination) are involved.
- When water runs over or through the ground surface, it may **pick up particles of soil**. This is noticeable in the water as **turbidity**.
- It also picks up particles of **organic matter and bacteria**.
- As surface water **seeps downward** into the soil and through the underlying material to the water table, most of the **suspended particles are filtered out**.
- This natural filtration may be **partially effective in removing bacteria** and other particulate materials.
- However, the **chemical characteristics** of the water may change and vary widely when it comes in contact with **mineral deposits**.
- As surface water seeps down to the water table, it dissolves some of the minerals contained in the soil and rocks.
- **Groundwater**, therefore, often contains **more dissolved minerals than surface water**.

Types of water

Pure water

Wholesome water

- **Water which is not chemically pure**; but **does not contain anything harmful to human health.**
- Will contain sufficient quantity of necessary elements or minerals.

Palatable water

- Water which is **aesthetically good looking and tasteful for drinking**

Potable water

- **Wholesome water + palatable water**

Contaminated water

Polluted water

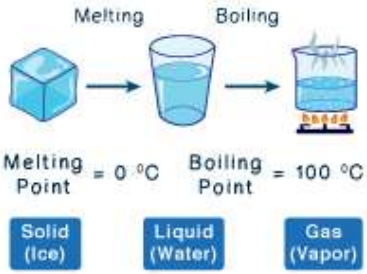
Water

Structure

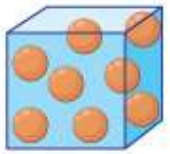


2 hydrogen atoms bond to oxygen atom, resulting in a bent shape

States of Matter



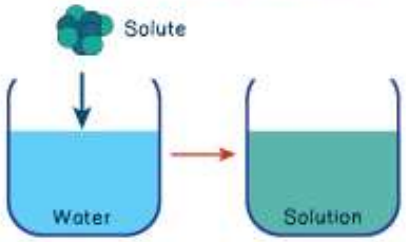
Density



	ρ (g/c³)
Solid	0.917
Liquid	0.997
Gas	0.6

The density is maximum (1 g/cm³) at 3.98 °C

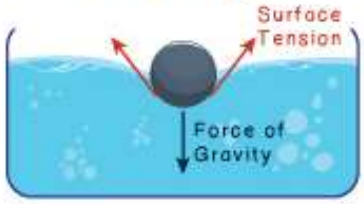
Universal Solvent



Capable of dissolving many substances

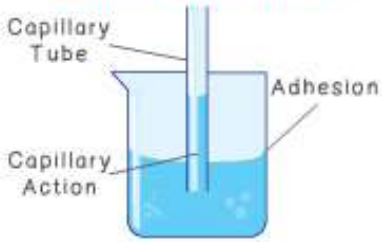
Surface Tension

$\sigma = 0.072\ Jm^{-2}$



Forms a film that can resist an external force

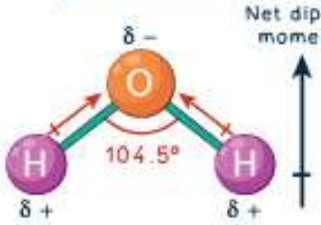
Capillary Action



Moves up a thin air column against gravity

Polarity & Dipole Moment

$\mu = 1.855\ D$



Polar molecule due to the asymmetric shape

Heat of Vaporization

$\Delta H_v = 2260\ J/g$



Specific Heat

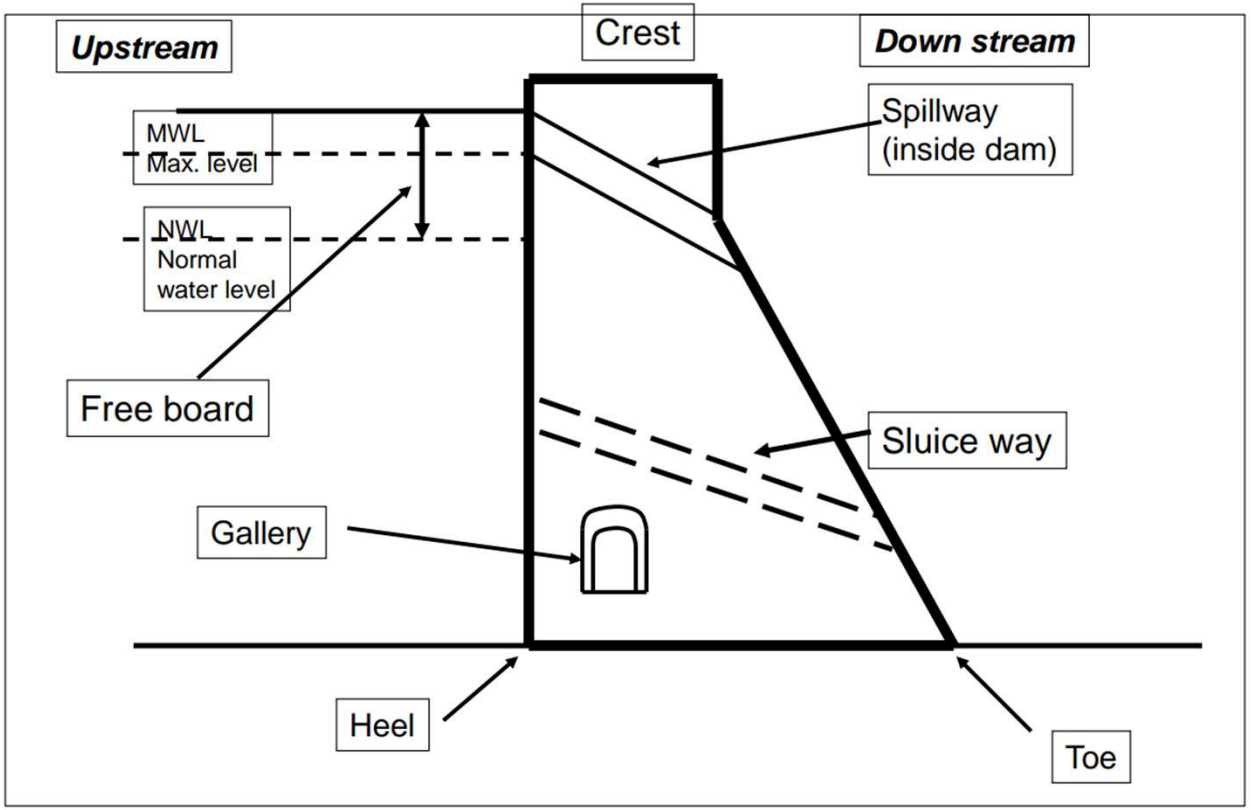
$c = 4.184\ J/g\text{-}°C$



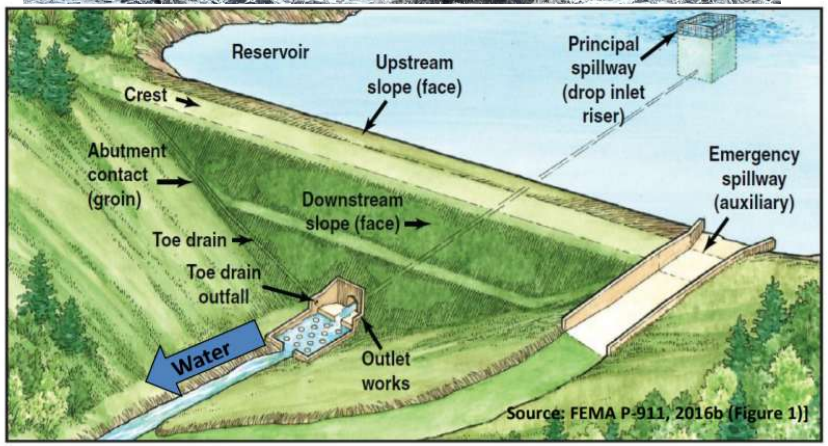
Dams

- Dam is a solid barrier constructed at a suitable location across a river valley to store flowing water.
- Storage of water is utilized for following objectives:
  - ✓Hydropower
  - ✓Irrigation
  - ✓Water for domestic consumption
  - ✓Drought and flood control
  - ✓For navigational facilities
  - ✓Other additional utilization is to develop fisheries

Dams


Upstream
Crest
Down stream
MWL
Max. level
Spillway
(inside dam)
NWL
Normal
water level
Free board
Sluice way
Gallery
Heel
Toe




Reservoir
Upstream
slope (face)
Principal
spillway
(drop inlet
riser)
Crest
Abutment
contact
(groin)
Emergency
spillway
(auxiliary)
Downstream
slope (face)
Toe drain
Toe drain
outfall
Water
Outlet
works
Source: FEMA P-911, 2016b (Figure 1)]

Components of Dam

- Heel: contact with the ground on the upstream side
- Toe: contact on the downstream side
- Abutment: Sides of the valley on which the structure of the dam rest
- Galleries: small rooms like structure left within the dam for checking operations.
- Diversion tunnel: Tunnels are constructed for diverting water before the construction of dam. This helps in keeping the river bed dry.
- Spillways: It is the arrangement near the top to release the excess water of the reservoir to downstream side
- Sluice way: An opening in the dam near the ground level, which is used to clear the silt accumulation in the reservoir side

Types of dams - Based on the function

- **Storage dams-**constructed to **store water** during the rainy season **when there is a large flow in the river**. Many small dams impound the spring runoff for later use in dry summers. Storage dams may also provide a water supply, or improved habitat for fish and wildlife. They may store water for **hydroelectric power generation, irrigation** or for a **flood control project**.
- **Diversion dams:** A diversion dam is constructed for the purpose of **diverting water** of the river into an off-taking canal (or a conduit). They provide **sufficient pressure for pushing water** into ditches, canals, or other conveyance systems. Such shorter dams are used for irrigation, and for diversion from a stream to a distant storage reservoir. A diversion dam is usually of low height and has a small storage reservoir on its upstream.
- **Detention dams:** Detention dams are constructed **for flood control**. A detention dam retards the flow in the river on its downstream during floods by storing some flood water.
- **Debris dams:** A debris dam is constructed to **retain debris such as sand, gravel, and drift wood** flowing in the river with water. The **water after passing** over a debris dam **is relatively clear**.
- **Coffer dams:** It is an enclosure constructed around the construction site to exclude water so that the construction can be done in dry. A cofferdam is thus a **temporary dam constructed for facilitating construction.**



Storage Arch Dam



Granite Reef Diversion Dam, USA



Mono Debris Dam, USA



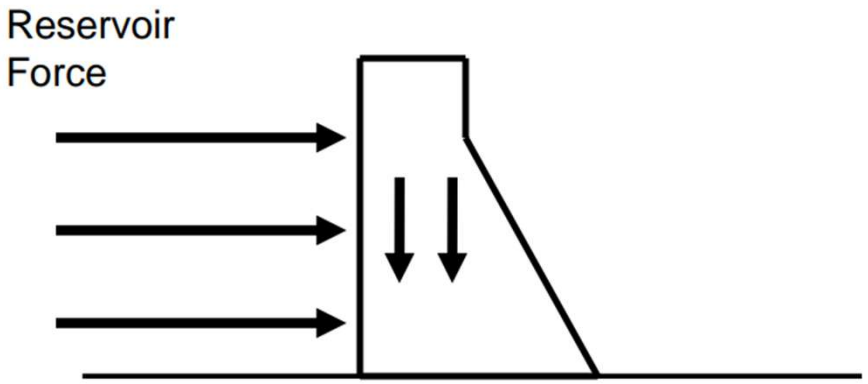
Coffer Dam

Based on structure and design

- **Gravity Dams:** A gravity dam is a massive sized dam fabricated from **concrete or stone masonry**. They are designed to **hold back large volumes of water**. By using concrete, the weight of the dam is actually able to resist the horizontal thrust of water pushing against it. This is why it is called a gravity dam.
- **Earth Dams:** An earth dam is made of earth (or soil) built up by compacting successive layers of earth, using the most impervious materials to form a core and placing more permeable substances on the upstream and downstream sides.
- **Arch Dams:** An arch dam is curved in plan, with its **convexity** towards the **upstream** side. An arch dam transfers the **water pressure and other forces mainly to the abutments by arch action**.
- **Buttress Dam:** Is a gravity dam reinforced by structural supports. Buttress is a support that transmits a force from a roof or wall to another supporting structure

Gravity Dams

- These dams are heavy and massive wall-like structures of concrete in which the whole weight acts vertically downwards.

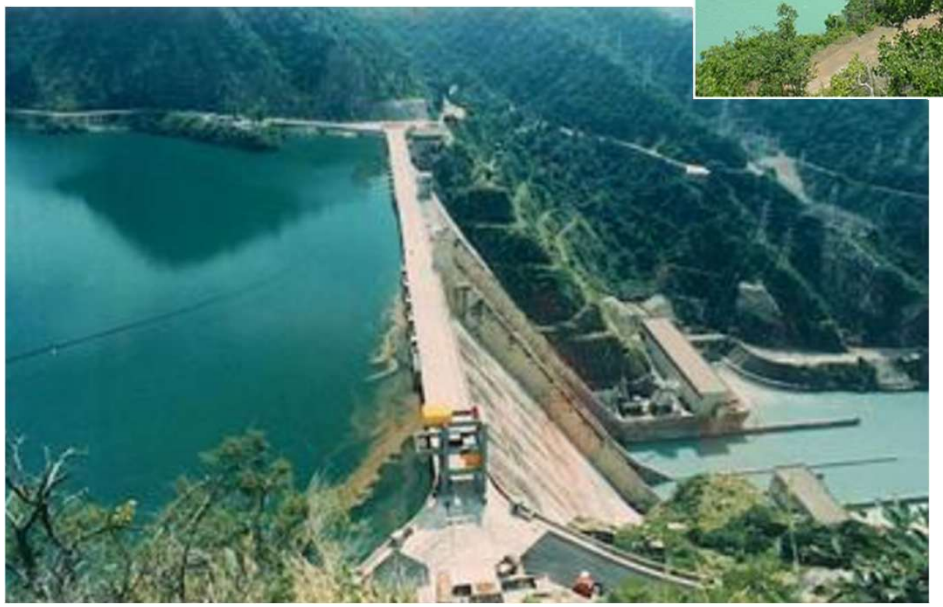


- As the **entire load** is transmitted **on the small area of foundation**, such dams are **constructed where rocks are competent and stable**.

Gravity Dams:



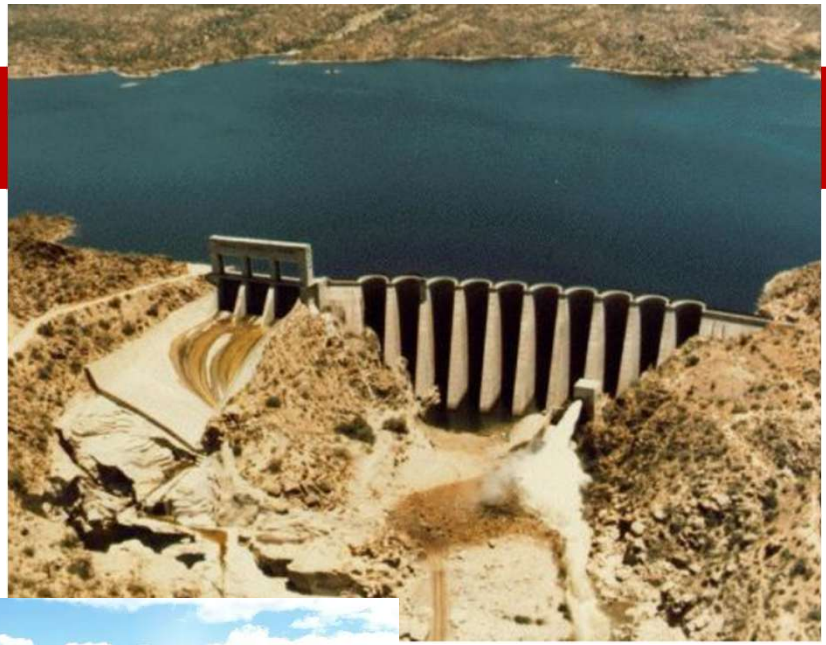
- Bhakra Dam is the highest Concrete Gravity dam in Asia and Second Highest in the world.
- Bhakra Dam is across river Sutlej in Himachal Pradesh
- The construction of this project was started in the year 1948 and was completed in 1963

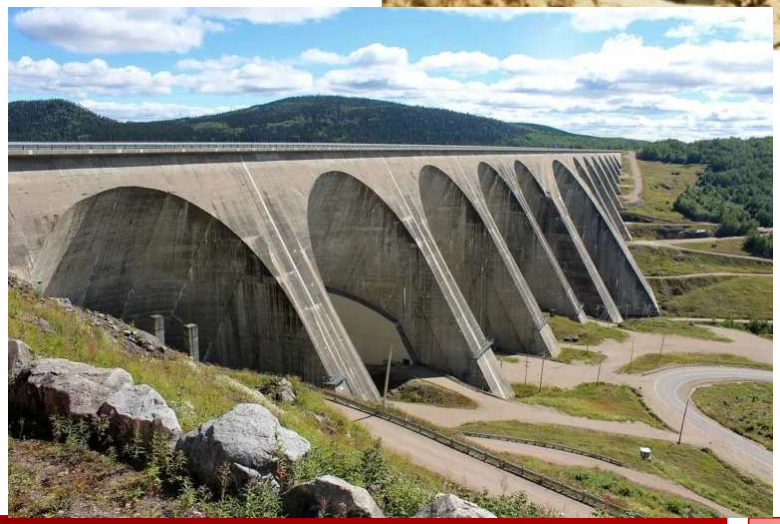


- It is 740 ft. high above the deepest foundation as straight concrete dam being more than three times the height of Qutab Minar.
- Length at top 518.16 m (1700 feet); Width at base 190.5 m (625 feet), and at the top is 9.14 m (30 feet)
- Bhakra Dam is the highest Concrete Gravity dam in Asia and Second Highest in the world.

Buttress Dam

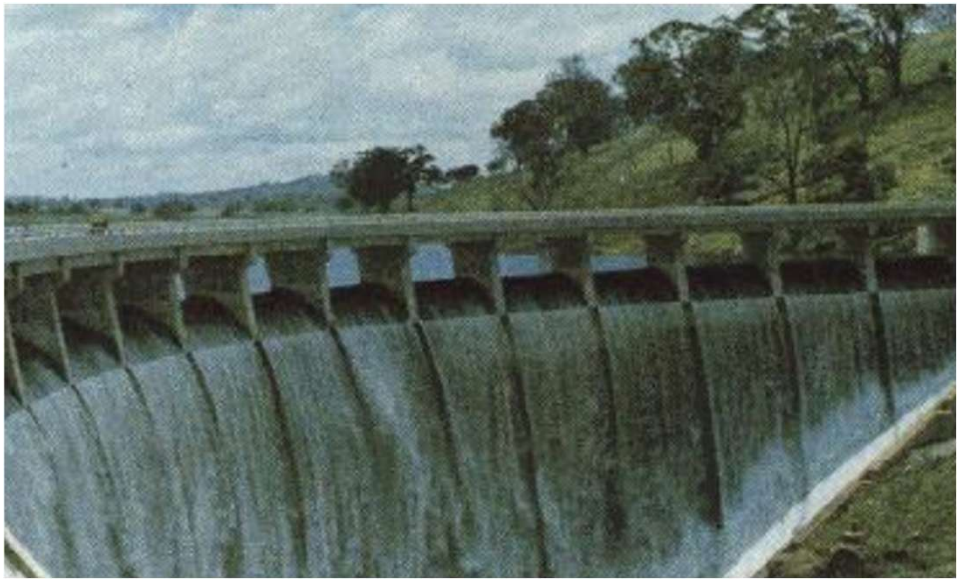
- Buttress Dam – Is a gravity dam **reinforced** by structural supports
- Buttress - **a support** that **transmits a force from a roof or wall to another supporting structure**
- This type of structure can be considered **even if the foundation rocks are little weaker**





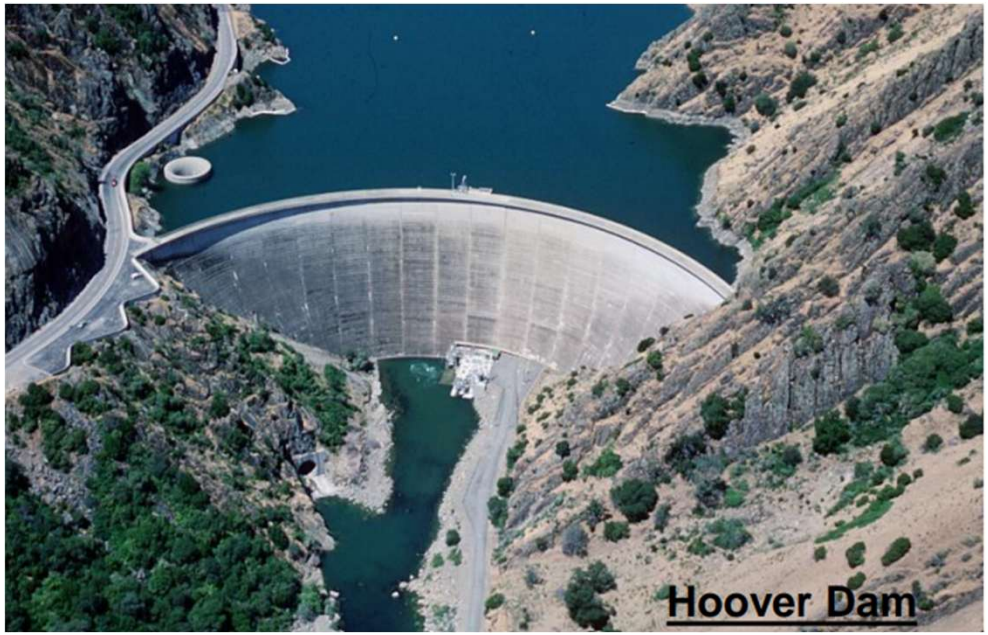
Arch Dams:

- These type of dams are concrete or masonry dams which are **curved or convex upstream** in plan
- This shape helps to transmit the major part of the **water load to the abutments**



- Arch dams are built across narrow, deep river gorges, but now in recent years they have been considered even for little wider valleys.

Arch Dam





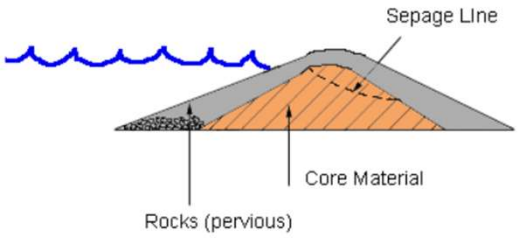
EMBANKMENT DAMS

EMBANKMENT DAMS (Rock Fill or Earth Fill Dams)

- They are mostly composed of natural materials such as, clay, sand, gravel etc...
- Impervious core is placed in the middle of the embankment body
- Generally riprap is used to control erosion

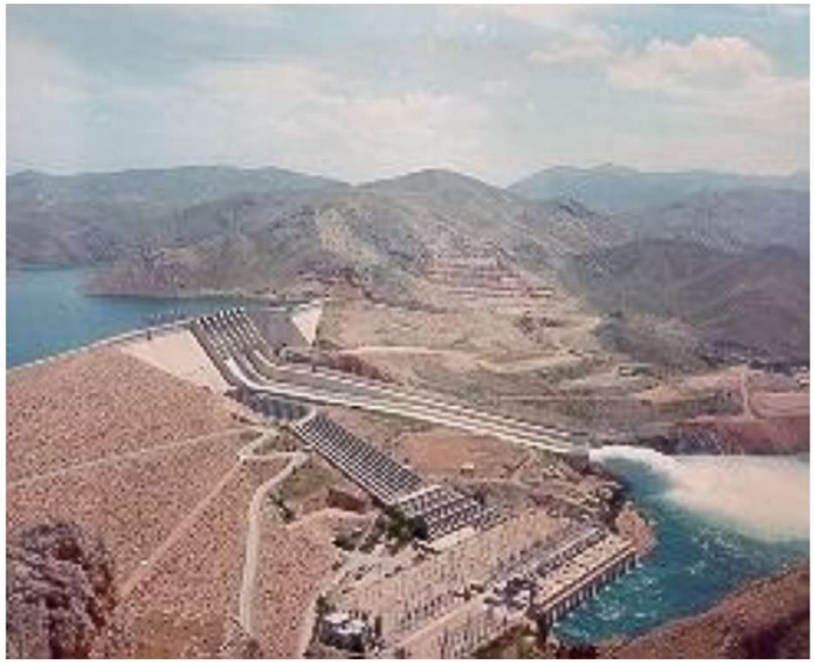






COMPOSITE DAMS

- Composite dams are combinations of one or more dam types.
- Most often a large section of a dam will be either an embankment or gravity dam, with the section responsible for power generation being a buttress or arch.



Gravity & Rock Fill

COFFER DAMS

- Temporary structures (sheet-pile, water-tight) that allow construction operations
- Diverts flow from construction areas until work completed





Water Quality Characteristics

- Physical Characteristics
  - Colour, Turbidity, Taste, Odour, Temperature
- Chemical Characteristics
  - pH, TDS, Total (Alkalinity, Hardness), Calcium, Magnesium, Fluoride, Chlorides, Sulphates, Nitrates, Total Nitrogen, Chloramines, Residual Chlorine, Iron, Ag, Cu, Zn
- Biological Characteristics
  - Pathogens

Examples



Evian 330ml Natural Mineral Water (Pack of 4)

by Evian

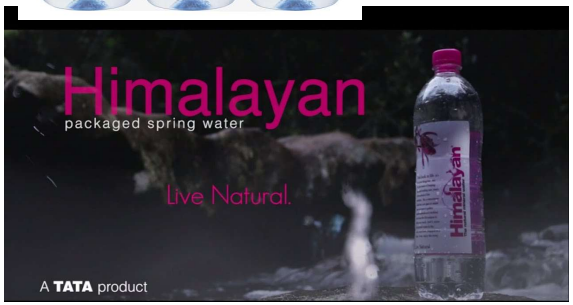
1 customer review

M.R.P.: ~~₹ 999.00~~

Price: ₹ 595.00 **FREE Delivery.** Details

You Save: ₹ 404.00 (40%)

- Pure Natural Mineral-Rich Water .
- No Chemicals Added .
- 100% Natural, 0% Processed .

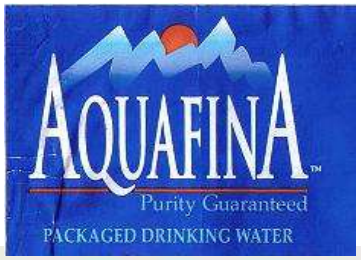


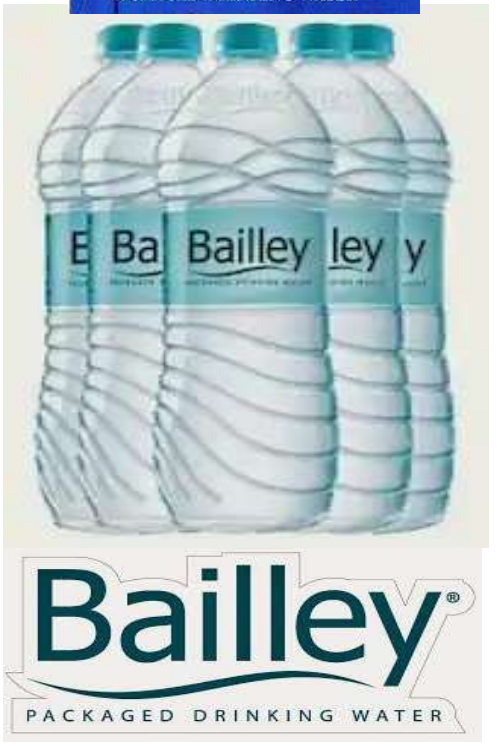
Himalayan Natural Mineral Water, which is available for Rs. 50/L

The water is bottled at source from a pure and pristine underground moving stream aquifer, which is about 400 feet below the surface, located at the foothills of the Shivalik range in the Himalayas. The catchment area has no human activity and is pollution-free. Himalayan is bottled in Dhaula Kuan, Himachal Pradesh, and available in 200 ml, 500 ml, 750 ml, 1 litre and 1.5 litre bottles.









Quality of Water (Testing of water)

- To know the extent of impurities present in water, based on this method and degree of treatment is decided.
- To ensure the quality of treated water before supply.

1. Physical Water Quality - Assessed by knowing the extend of physical impurities In water. Physical water quality parameters are those which respond to human senses.

Physical Water Quality

Temperature: (10° C to 20 ° C)

1. Warm temperature is ideal for microbial growth.
2. Solubility of gases dependent of temperature. Higher the temperature, lower will be solubility.
3. Chemical reactions are temperature controlled.

Colour: (5 TCU to 20 TCU)

1. Color in drinking-water may be due to the presence of organic matter such as Humic Substances, metals such as Iron and Manganese, or highly colored industrial wastes
2. Term ‘color’ is used to mean True Color – Due to Dissolved Solids
3. Term ‘Apparent Color’ includes not only the color due to substances in solution but also that due to Suspended Matter or Colloid Solids.
4. Colour in water measured on Platinum scale and expressed in terms of **True colour Units (TCU)** by using device known as **Tinto meter**.

Physical Water Quality

Temperature: (10° C to 20 ° C)

1. Warm temperature is ideal for microbial growth.
2. Solubility of gases dependent of temperature. Higher the temperature, lower will be solubility.
3. Chemical reactions are temperature controlled.

Colour: (5 TCU to 20 TCU)

1. Color in drinking-water may be due to the presence of organic matter such as Humic Substances, metals such as Iron and Manganese, or highly colored industrial wastes
2. Term ‘color’ is used to mean True Color – Due to Dissolved Solids
3. Term ‘Apparent Color’ includes not only the color due to substances in solution but also that due to Suspended Matter or Colloid Solids.
4. Colour in water measured on Platinum scale and expressed in terms of **True colour Units (TCU)** by using device known as **Tinto meter**.

Physical Water Quality

3. Turbidity (5 TU to 10 TU)

- Opaqueness in water is known as turbidity. It is the resistance to passage of light through water, by the particles.
- Turbidity in water is caused by suspended and colloidal particles.
- Dissolved particles do not possess any surface area. They go into molecular level. So turbidity is offered by suspended and colloidal particles.
- Turbidity is measured in 'Silica Scale' and expressed in terms of “Turbidity Units (TU)”
- Turbidity is measured by
- (i) Devices working on the principle of light absorption - Turbidimeters.
- (ii) Device working on the principle of light scattering - Nephlometer.

Turbidity is measured using:

- Turbidity rod
- Jacksons turbidimeter
- Baylis turbidimeter
- Nephelometer





Physical Water Quality

Odour and taste (< 3 TON)

- Odour in water is measured by a device known "Osmoscope". It is an inhaler with capacity 200 mL.
- To ensure the safety of the person who conducts test, diluted samples always used.
- Expressed in terms of Threshold Odour Number (TON).
- There are no such tests to measure taste. Generally, a sample with bad odour is expected to taste bad.
- FTN - Flavoured Threshold Number.

Physical Water Quality

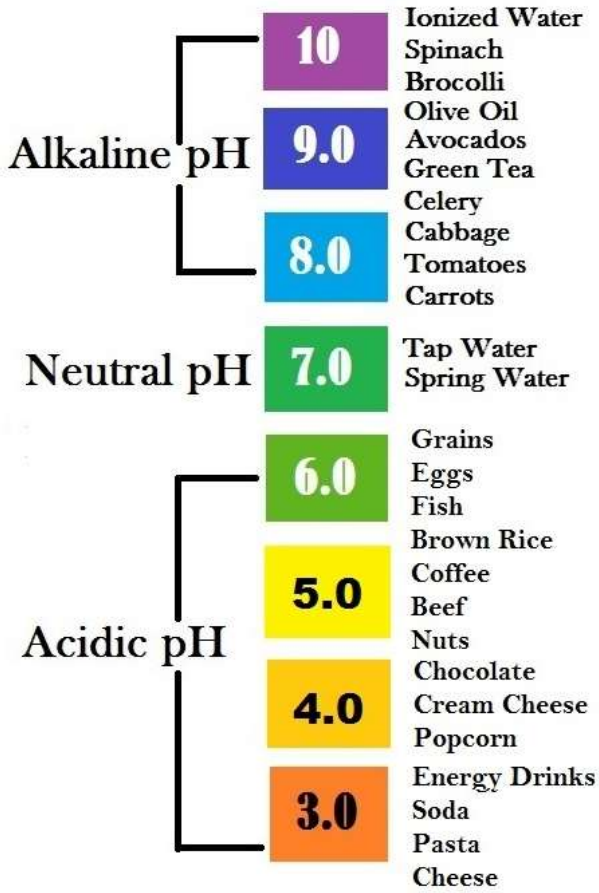
pH

- Universally to express the intensity of the acid or alkaline condition of a solution
- Expressed on a scale ranging from 0 to 14
- Logrithm of Reciprocal of Hydrogen ion activity (moles/L)
- Recommended pH range for treated drinking waters is 6.5 to 8.5

$$pH = -log\{H^{+}\}\, or$$

$$pH = \log\{\frac{1}{H^{+}}\}$$

- pH is usually represented from 0 to 14, with pH 7 at 25°C representing absolute neutrality



Chemical Water Quality

Solids

- Refer to matter suspended or dissolved in water or wastewater
- Usual definition of "solids", however, refers to the matter that remains as residue upon evaporation and drying (103 to 105 °C)*
- Total Solids in a liquid sample consist of Total Dissolved Solids (TDS) and Total Suspended Solids (TSS)
- **Total Solids :** A well-mixed sample (100 mL) is evaporated in a weighed dish and dried to constant weight in an oven at 103 to 105°C for 1 hour

Solids (Dissolved, Suspended)

- **Total Dissolved Solids :** A well-mixed water sample (100 mL) is filtered through a standard glass fiber filter (2.0 µm pore size) , and the **filtrate** is evaporated to dryness in a weighed dish and dried to constant weight at **180°C** for 1 hour
  - TDS↑ Hardness ↑

- **Total Suspended Solids:** A well-mixed water sample (100 mL) is filtered through a weighed standard glass-fiber filter (2.0 µm pore size) and the **residue retained on the filter** is dried to a constant weight at **103 - 105°C** for 1 hour

Solids Standards

Indian Standards	Desirable Limit (mg/L)	Permissible Limit in Absence of Alternate Sources
IS 10500:2012	500	2000

Environmental Significance

- Laxative or reverse effects on people whose bodies are not used to the higher levels
- Imparts taste to water
- Water stain glassware

Iron & Manganese

- Groundwaters that contain appreciable amounts of Fe or Mn or both are always devoid of dissolved oxygen and are high in CO_2 content.
- Both Fe & Mn interfere with laundering operations, impart **objectionable stains** to plumbing fixtures, and cause difficulties in distribution systems by supporting growths of **iron bacteria**

Iron bacteria are small living organisms which naturally occur in soil, shallow groundwater, and surface waters. These nuisance **bacteria combine Fe (or Mn) and oxygen** to form deposits of **"rust,"** bacterial cells, and a **slimy material that sticks the bacteria to well pipes**, pumps, and plumbing fixtures. The bacteria are **not known to cause disease**, but can cause ***undesirable stains, tastes and odors***





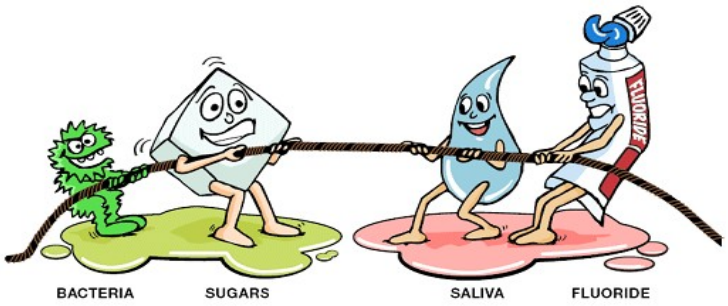
Iron & Manganese

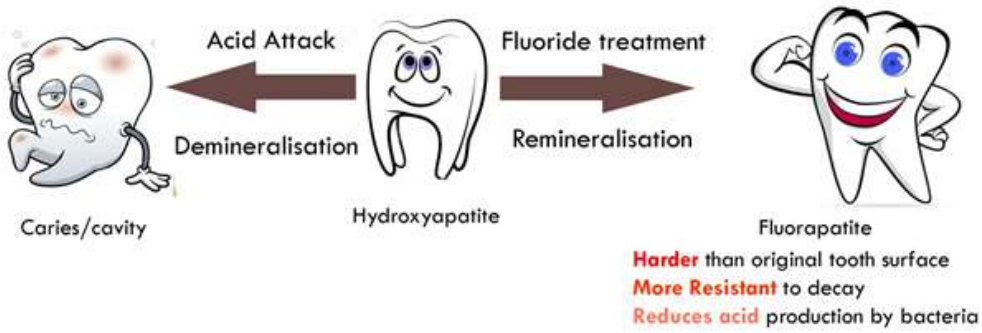
Indian Standards	Desirable Limit (Fe) (mg/L)	Permissible Limit in Absence of Alternate Sources
IS 10500:2012	0.3	No relaxation

Indian Standards	Desirable Limit (Mn) (mg/L)	Permissible Limit in Absence of Alternate Sources
IS 10500:2012	0.1	0.3

Fluoride

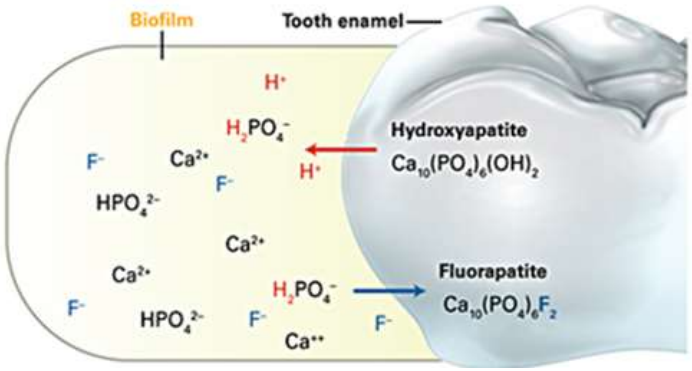
- The discovery of high consumption of Fluoride (two-edge sword) being harmful for humans and animals was made in India in 1937 (Nalgonda-Telangana)
- When present in less concentration of < (0.8-1.0 mg/L) – Cause ***Dental Caries*** (*Tooth Decay*)
- Fluoride makes the tooth more resistant to acid attack from bacteria.
- When fluoride compounds are in your mouth, they can actually make teeth stronger and prevent cavities.





Fluoride

- Higher Fluoride Concentration (1.5-2.0 mg/L) cause Discoloration of teeth (***Dental Fluorosis***)
- Dental fluorosis : Tooth becoming colored from yellow to brown to **black**
- At still higher concentration, (> 3.0 mg/L) **Skeletal Fluorosis** which affects the bone and ligaments
- Fluorosis is a crippling disease resulted from deposition of fluorides in the hard and soft tissues of body





Dental Fluorosis



Skeletal Fluorosis

Chemical Water Quality

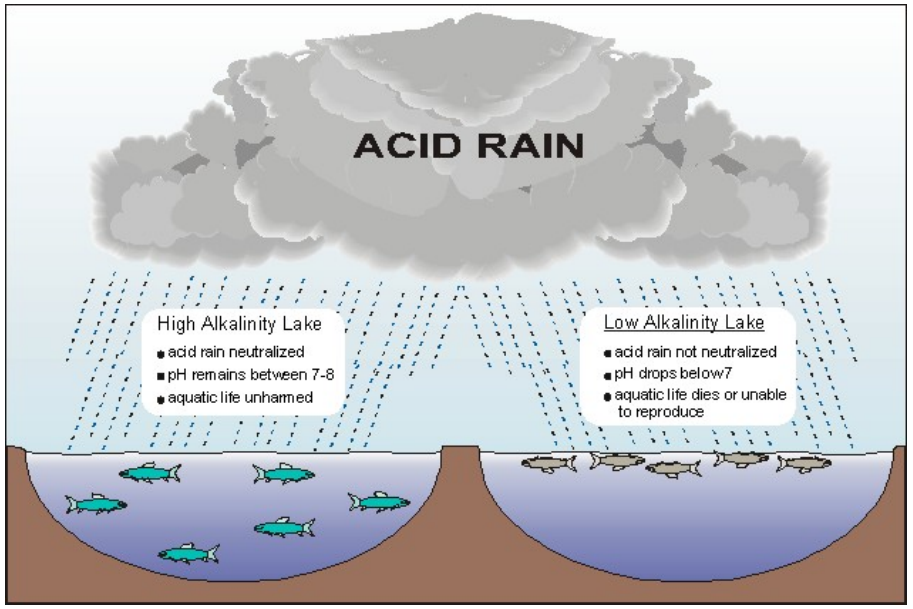
Fluoride - Limits

Indian Standards	Desirable Limit (F) (mg/L)	Permissible Limit in Absence of Alternate Sources
IS 10500:2012	1.0	1.5

Alkalinity

- Alkalinity of a water is a measure of its capacity to **Neutralize Acids**
- Alkalinity in natural water is due to:

1. **Salts of Weak Acids**
    - Carbonate, bicarbonate
2. **Weak or Strong Bases**
    - Ammonia
    - Hydroxides

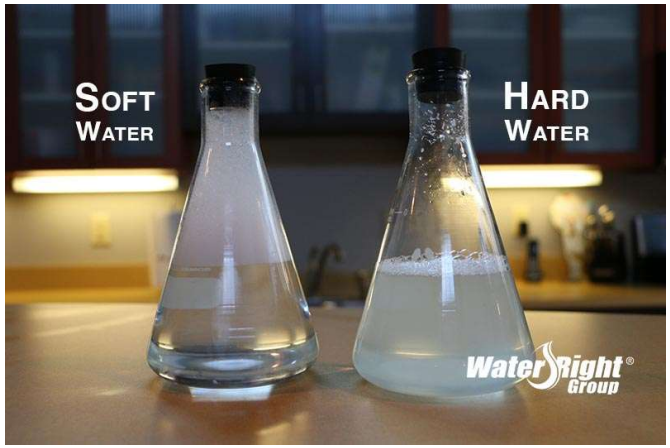


Alkalinity Data - Application

- **Chemical Coagulation**
  - Chemicals used for coagulation of water and wastewater react with water **to form insoluble Hydroxide precipitates**. The H^+ ions released react with the alkalinity of the water. Thus, the alkalinity helps the coagulant to be effective.
- **Water Softening**
  - Alkalinity - major item to be considered in calculating lime and soda ash requirements in softening of water.
- **Corrosion Control**
  - Used to calculate the degree of saturation of Calcium Carbonate in water.

Indian Standards	Desirable Limit (mg/L)	Permissible Limit in Absence of Alternate Sources
IS 10500:2012	200	600

Hardness

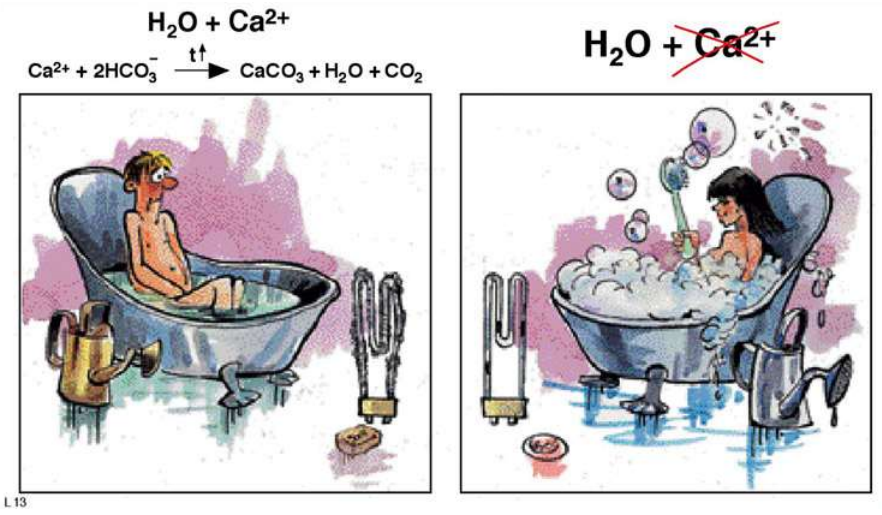


- ***Soft Water -*** Forms lather easily with soap
- **Hard water -** Needs more soap to form lather
  - Because of dissolved chemicals in the hard water react with soap to form a scum.

Types of Hardness

1. **<u>Temporary Hardness</u>** (Easily removed by boiling)
   Due to the presence of HCO_3^- of Ca^{2+} & Mg^{2+}
2. **<u>Permanent Hardness</u>** (Cannot be removed by boiling)
   Due to the presence of Cl^- & SO_4^{2-} of Ca^{2+} & Mg^{2+}

Indian Standards	Desirable Limit (mg/L)	Permissible Limit in Absence of Alternate Sources
IS 10500:2012	200	600



Chlorides - Significance

- Too much of Cl^- in soil, can accumulate in leaf tissue, resulting in leaves with a scorched or burned appearance.
- Evapotranspiration tends to increase Cl^- and salinity at the root zone of irrigated plants, making it difficult for crops to take up water.
- Interferes in COD Analysis



Indian Standards	Desirable Limit (mg/L)	Permissible Limit in Absence of Alternate Sources
IS 10500:2012	250	1000

Toxic Chemicals (Heavy Metals)

(i) Damage central nervous system
(i) Damage blood cells.
(ii) Damage blood vessels
(iv) cancerogenic (carcinogenic)

Toxic Chemical Substances	Drinking water Standards.
Lead	< 0.1 mg/L
Arsenic	<0.05 mg/L
Barium	< 1 mg/L
Boron	<0.01 mg/L
Cadmium	<0.01 mg/L
Chromium	<0.05 mg/L
Phenols	<0.001 mg/L
Mercury	<0.001 mg/L.
Cyanide.	<0.05 mg/L

Biological Water Quality

- Microbial Examination of Water
- Indicator – **E-coli**, Most probable number (MPN<1no/100 mL)
- E-coli organism belongs to Coliform group of bacteria.
- It is non pathogen.
- E-coli organism is present in intestines of warm blooded animals including humans.
- They are responsible for converting food into energy and waste products.
- But E-coli organism is treated organism. Presence of E-coli indicates, presence of pathogens.

Water Borne Diseases :

Name of the disease.	Organism causing the disease	Category.
1. Cholera	Vibrio cholerae or Vibrio comma.	Bacteria.
2. Typhoid.	Salmonella typhi	Bacteria.
3. Paratyphoid.	Salmonella paratyphi	Bacteria.
4. Dysentry	Shigella dysentena.	Bacteria.
5. Amebiasis (intestine)	Entamoeba & Hystolytica.	Protozoa.
6. Giadiasys (intestine)	Giardia Lamblia.	Protozoa.
7. Jaundice (liver)	Infectious Hepatitis	Virus
8. Polio.	Polio myelitis	Virus.

Indian Standard for Drinking Water as per BIS specifications (IS 10500-2012)

S.NO.	Parameter	Requirement desirable Limit	Remarks
1.	Colour	5	May be extended up to 50 if toxic substances are suspected
2.	Turbidity	10	May be relaxed up to 25 in the absence of alternate
3.	pH	6.5 to 8.5	May be relaxed up to 9.2 in the absence
4.	Total Hardness	300	May be extended up to 600
5.	Calcium as Ca	75	May be extended up to 200
6.	Magnesium as Mg	30	May be extended up to 100
7.	Copper as Cu	0.05	May be relaxed up to 1.5
8.	Iron	0.3	May be extended up to 1
9.	Manganese	0.1	May be extended up to 0.5
10.	Chlorides	250	May be extended up to 1000
11.	Sulphates	150	May be extended up to 400
12.	Nitrates	45	No relaxation
13.	Fluoride	0.6 to 1.2	If the limit is below 0.6 water should be rejected, Max. Limit is extended to 1.5
14.	Phenols	0.001	May be relaxed up to 0.002
15.	Mercury	0.001	No relaxation
16.	Cadmium	0.01	No relaxation
17.	Selenium	0.01	No relaxation
18.	Arsenic	0.05	No relaxation
19.	Cyanide	0.05	No relaxation
20.	Lead	0.1	No relaxation
21.	Zinc	5.0	May be extended up to 10.0
22.	Anionic detergents (MBAS)	0.2	May be relaxed up to 1
23.	Chromium as Cr^{+6}	0.05	No relaxation
24.	Poly nuclear aromatic Hydrocarbons	--	--
25.	Mineral Oil	0.01	May be relaxed up to 0.03
26.	Residual free Chlorine	0.2	Applicable only when water is chlorinated
27.	Pesticides	Absent	--
28.	Radio active	--	--

Activity 3 Guess the pH of…….

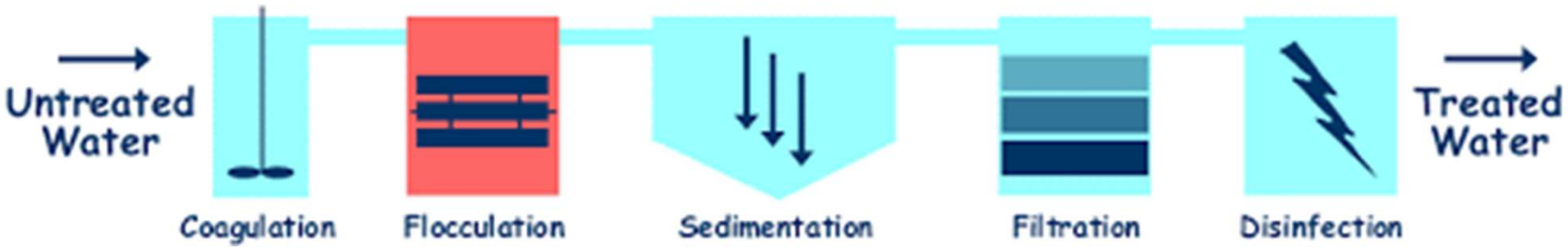


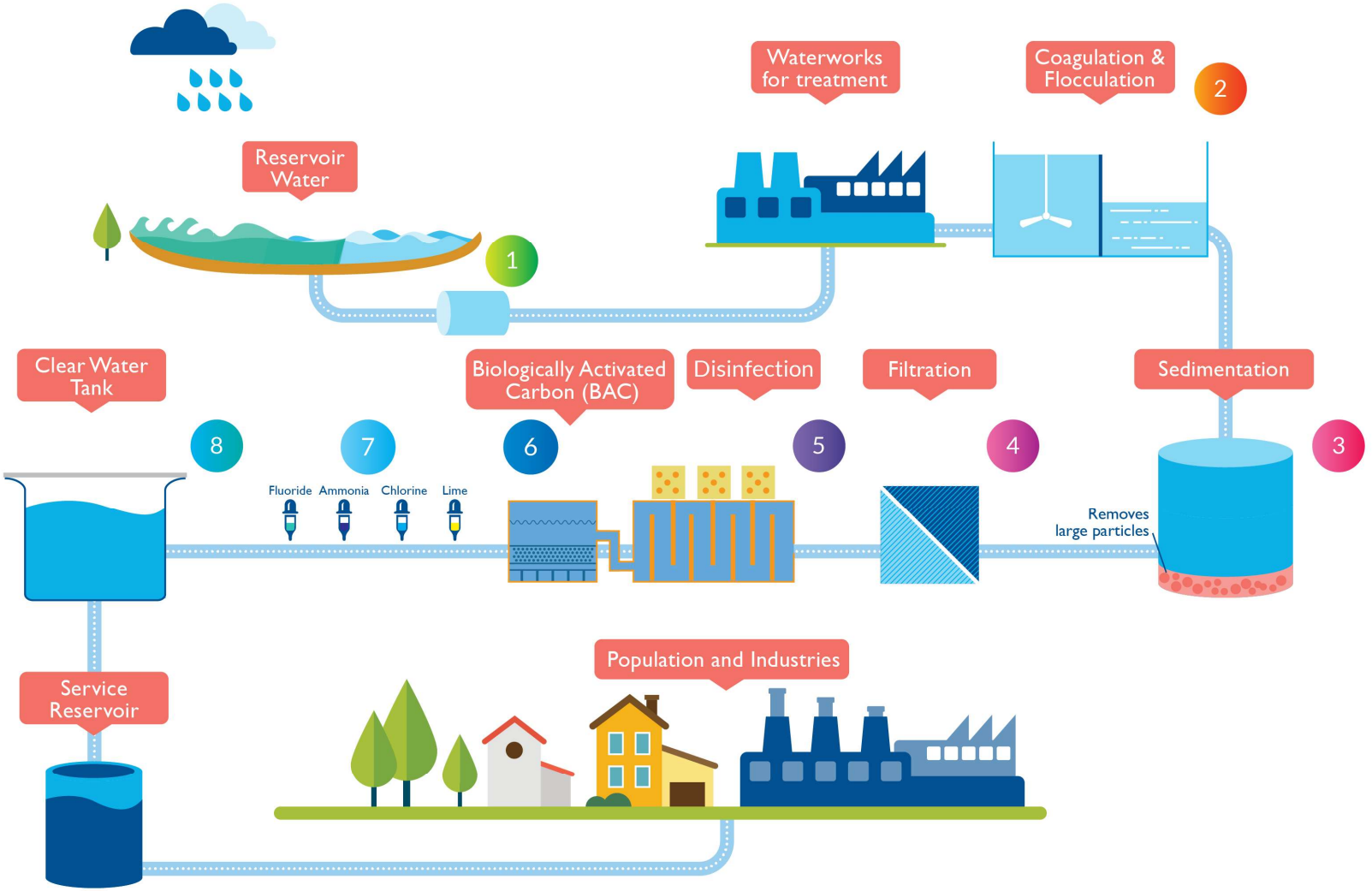




Water treatment

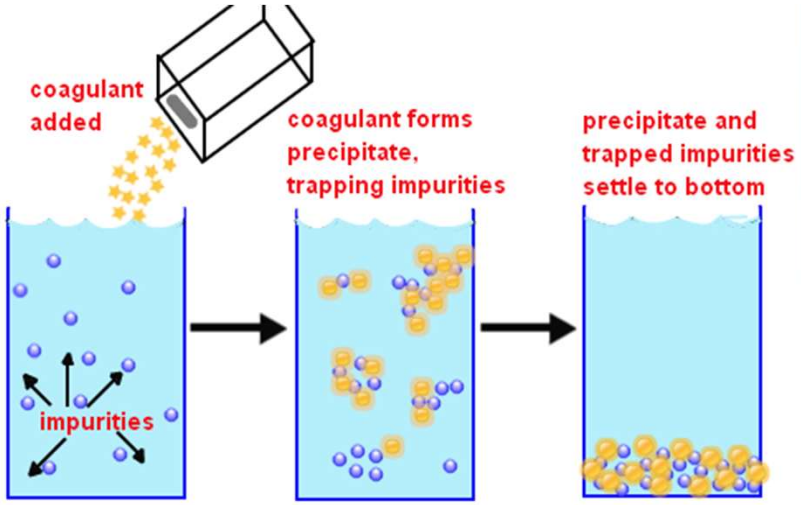
- Water treatment is any process that improves the quality of water to make it appropriate for a specific end-use.
- The end use may be drinking, industrial water supply, irrigation, river flow maintenance, water recreation or many other uses, including being safely returned to the environment.

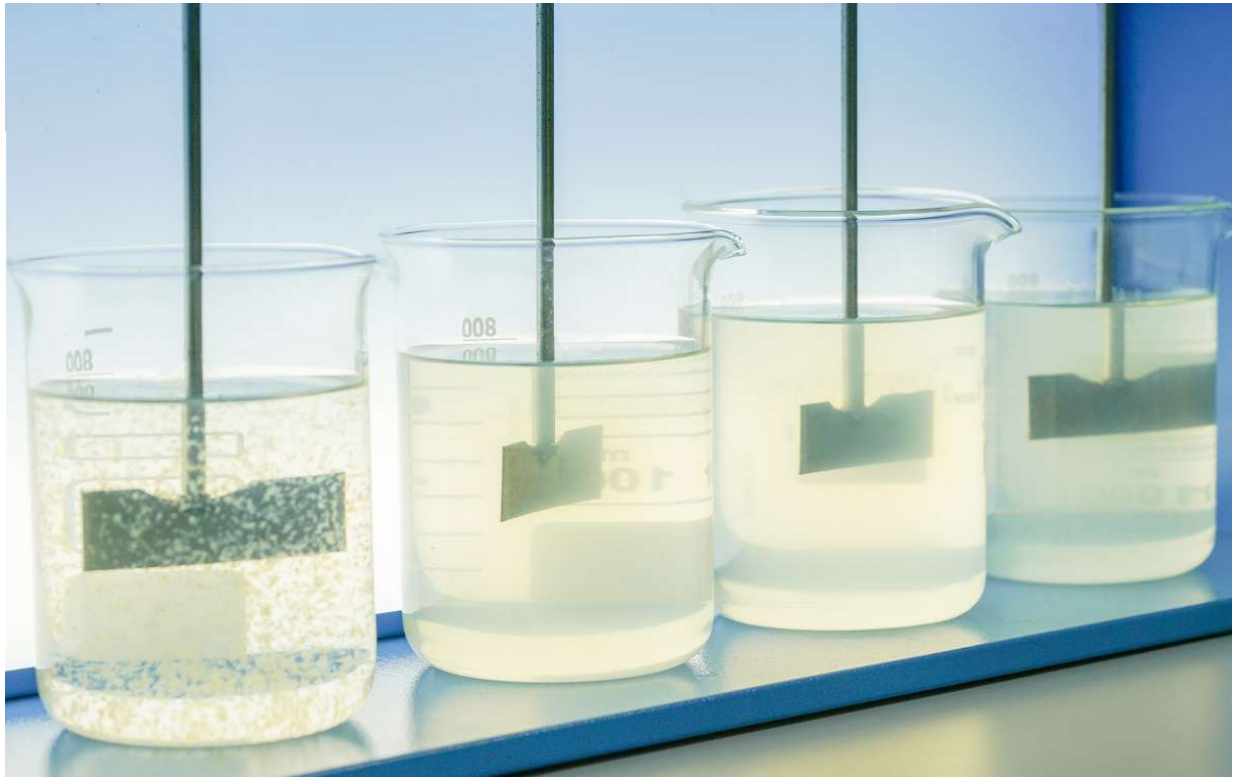




Typical Surface Water Treatment Plant

Coagulation





Wastewater

- Water that has been used in the home, in a business, or as part of an industrial process.
- Water that has been used in washing, flushing, manufacturing etc.



Interesting stories



Stamped Edition, 6d.

THE ILLUSTRATED LONDON NEWS

ILLUSTRATED LONDON NEWS ONE PENNY NEWSPAPER

No. 928.—VOL. XXXIII.] SATURDAY, JULY 24, 1858. [WITH A SUPPLEMENT, FIVEPENCE

THE PURIFICATION OF THE THAMES.

It is an old experience, public as well as private, that the longer a right thing is allowed to remain undone the more difficult it becomes to do it. The great question of the purification—or rather of the non-pollution—of the Thames is the newest and most flagrant example of this ancient truth. Error has been permitted to accumulate upon error, neglect upon neglect, and wrong upon wrong, until the evil consequences, strengthened and complicated by the lapse of time, have become so unmanageable that action or inaction is equally dangerous. The Thames, which, fifty years ago, ran through London in a clear and limpid stream, over whose current it was a pleasure to be rowed, in whose waves it was delightful to bathe, and of whose pure waters it was wholesome to drink, has, by sheer neglect on the part both of the people and the Government, become a foul sewer, a river of pollution, a Stream of Death, festering and reeking with all abominable smells, and threatening three millions of people with pestilence as the penalty of their ignorance and apathy. It was the duty as well as the interest of London to keep its noble river, the source of all its wealth and much of its beauty, as clear as Nature gave it. But the initial difficulty in the case was that there was no London that could undertake the work. There was an old and small city, with rights and powers of self-government, surrounded by a congeries of towns, boroughs, and villages, larger than itself, and growing larger every day, all of which were equally interested in this great achievement, but none of which had the means of taking a step for the furtherance of the common design. The city or municipality of London—the mere nucleus of that mightier conglomeration of cities which form the actual metropolis of the British Empire—was not likely to tax the dwellers within its own narrow and defined area with the cost of the sewerage of the whole metropolis; and it had no power to levy a shilling for the purpose on the people of Westminster, Finsbury, Marylebone, or any other outlying borough. The metropolitans, as distinguished from the mere Londoners of the City proper, had not the spirit or the common sense to perceive that they were as much entitled to a municipal organisation as any other assemblage of British subjects living in towns and boroughs. A few persons may have urged the necessity of reforming the old and corrupt Municipality, which dated from the days of the Plantagenets and Tudors, and including within its amended jurisdiction the whole mighty metropolis of the days of Victoria; but their efforts met with no adequate support; and the Imperial Government, that never does anything except on compulsion, looked on and said nothing. If successive Governments had any feeling at all upon the subject it was one of satisfaction at the apathy which the people displayed; for Home Secretaries, and Foreign Secretaries, and the whole posse of public functionaries in Downing-street and Whitehall, entertain a stupid and we might call it unconstitutional jealousy of the power that might be exercised by so great a Municipality; and fear that the Lord Mayors of London in the nineteenth century may become as formidable to the kingly and aristocratic power as their predecessors were in the time of Richard II. and III. Many evils have resulted from this state of things;—evils of omission and of commission, of jurisdiction and of no jurisdiction, of bad government and of no government, of ill drainage and of no drainage. Year after year these evils have gone on augmenting; and, after the true British fashion, would have gone on augmenting for half a century longer, or for some uncountable period



TOXIC BAIT AFLOAT ON THAMES OF JALPAIGURI

Down the river, flotilla of dead fish

OUR CORRESPONDENT

Jalpaiguri, Nov. 28: Not all dawns break like this for Mamata Das. Swept out of bed at 6am by ripples of excitement swirling outside her home, Mamata found herself wading into the Karala, the "Thames of Jalpaiguri".

What else will you do when you woke up to news that lifeless fish after fish are bobbing up the river?

They floated down in all shapes, sizes and species — the boal, a catfish and a gourmet's delight in north Bengal, *tangra*, *ark*, American rohu, byne and puti. Some weighed 3 kilos, some 12.

"As soon as I heard about the fish, I went to the river at 6am and began collecting the fish. I took some for us and gave some to others," said Mamata, a homemaker whose husband works as a construction labourer. Then came the daylight knock. Alarmed that the fish could have died because of some toxic substance, officials raided homes and seized the catch, some from boiling pots and frying pans.

Among those who splashed into the river, called the local Thames because it bifurcates Jalpaiguri, were the sons of Harimohan Yadav, a day labourer.

The pieces of the *boal* his sons had gathered had just gone into the pot when the civic officials, alerted by a disaster management team, came calling. "We hardly get the luxury to taste fish like *boal* and my sons had bought a fair-sized one," Harimohan said. A kilo of large *boal* costs around Rs 400 and the smaller one Rs 200.

However, wiser counsel prevailed. "When the officials said that we could fall ill, I handed over the fish," Harimohan added. Similar scenes unfolded in several houses along a 5km stretch of the embankment in the heart of Jalpaiguri, the poor neighbourhood nursing disappointment at the slip between the pot and the lip but some realising that they may have been pulled back from the jaws of a health disaster.

The cause behind the mass death is yet to be established. Officials felt that someone could have thrown "some toxic chemical" into the river by accident. The officials added that they were at a loss on the motive if someone did so deliberately.

A pollution control board official said the pH level of the water indicated it had turned acidic because of contamination. The level of pH indicates hydrogen ion activity in a solution — the lower the level, the higher the acidity.

"The pH level of safe water is usually around 7.2 but I have found it to be below 4 that indicates that the river water has been contaminated either by poison or by some chemical. Further tests in our laboratory will confirm the actual source," said Sukriti Lama, a pollution control scientist who arrived from Siliguri and collected samples.

People scoop up dead fish from the Karala river in Jalpaiguri on Monday. Picture by Biplab Basak

Cooum River - Chennai





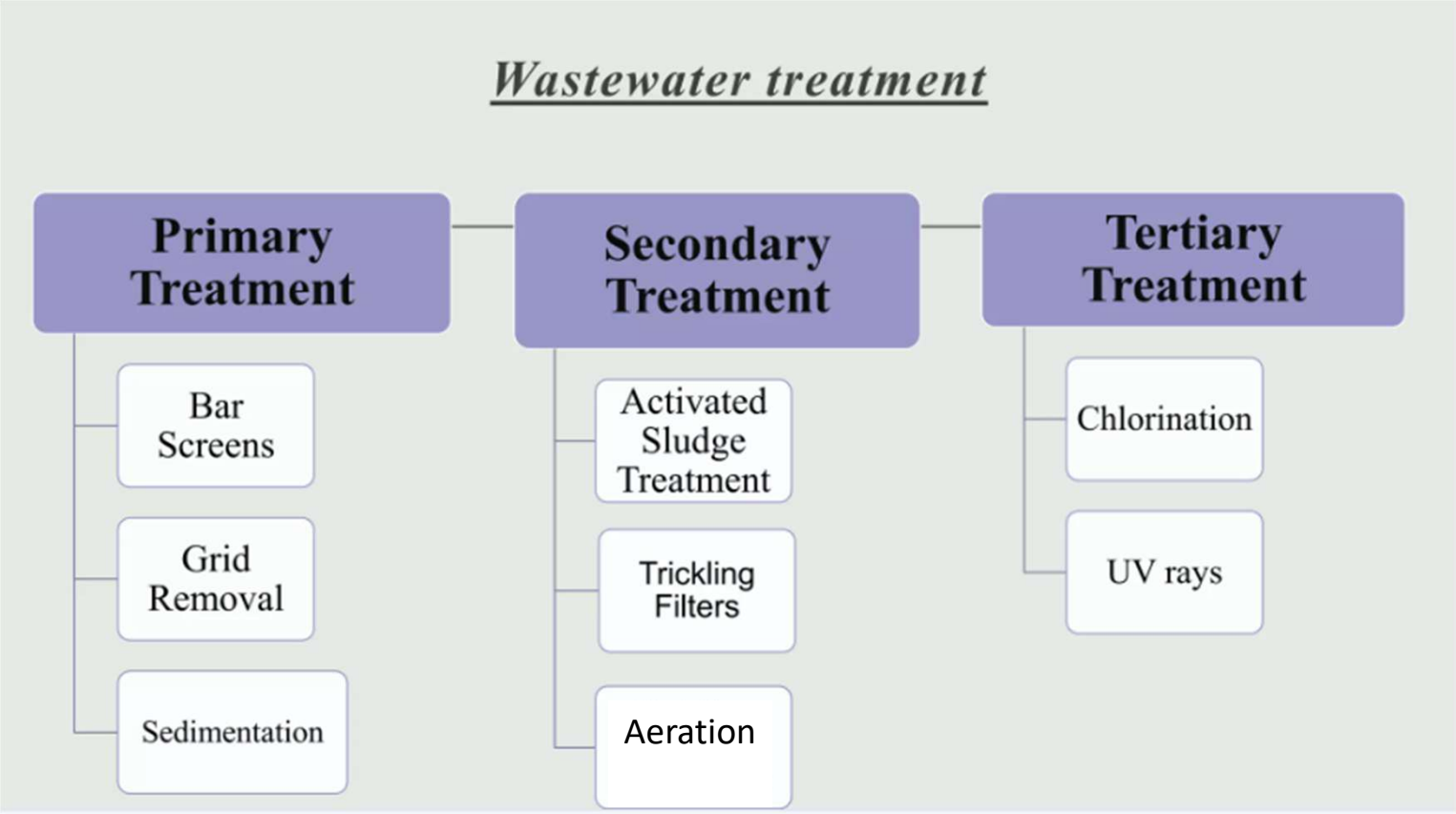
Source & Sewage composition

- In the form of Physical impurities, dissolved chemical impurities & Microbiological impurities
  - Solids like vegetable peelings, rice, plastic covers, etc.
  - Oil & Grease from Kitchen / Laundry
  - Suspended solids from kitchen
  - Chemicals from detergents
  - Mild colors from laundry / food
  - Pathogenic microbes from fecal matter

Wastewater Composition = 99.95% Water + 0.05% Impurities

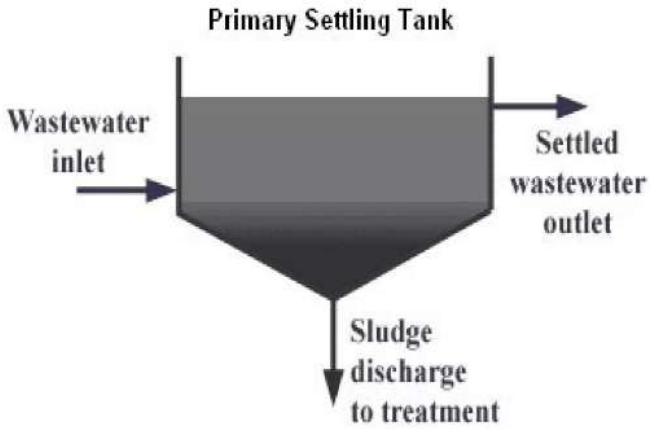
Levels of Treatment

Primary treatment	Secondary treatment	Tertiary (advanced)
• Removal by physical separation of grit and large objects (material to landfill for disposal)	• Degrade the biological content of the waste through aerobic biological processes. • Lowers suspended solids content	• Raise the quality of the water to domestic and industrial standards, or to meet specific requirements around the safe discharge of water. • In the case of water treated by municipalities, tertiary treatment also involves the removal of pathogens, which ensures that water is safe for drinking purposes.


Wastewater treatment
Primary Treatment
Bar Screens
Grid Removal
Sedimentation
Secondary Treatment
Activated Sludge Treatment
Trickling Filters
Aeration
Tertiary Treatment
Chlorination
UV rays

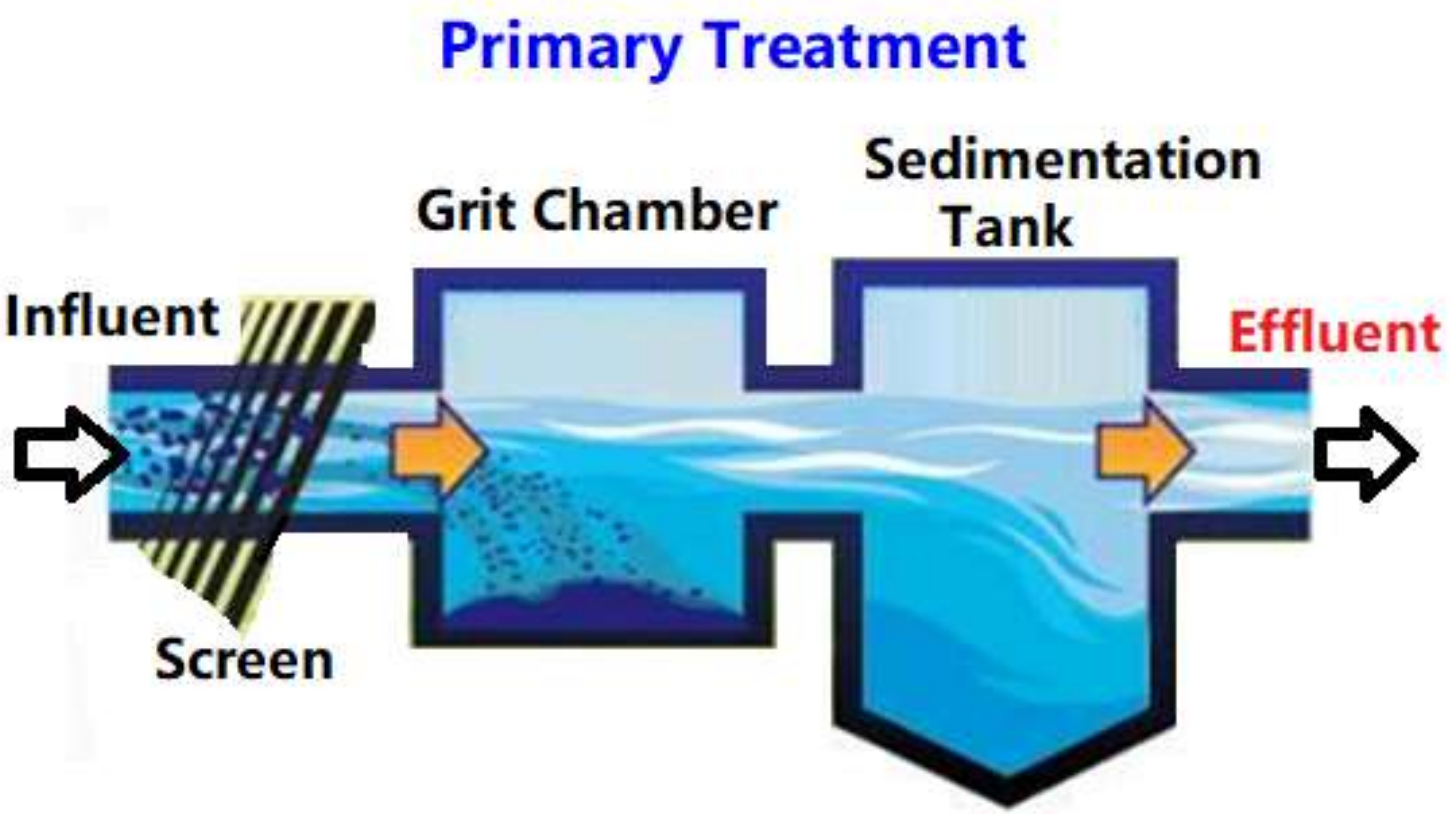
Primary Treatment

- **Methods used in primary treatment**
- Bar screens
- Grit Chamber
- Sedimentation Tank - primary Settling tank



Primary wastewater treatment is a physical process that removes floating and settleable materials from wastewater.

Primary Treatment



Primary Treatment

Screening

- Screens made of narrow metal bars/screens that block floating debris like wood, bags and very large particles.

Grit removal

- **Grit chambers** are long, narrow tanks that slow down the flow of wastewater to allow smaller solid particles like sand, eggshells, and coffee grounds to settle out.

Sedimentation

- Sedimentation tanks, also known as primary clarifiers, allow suspended solids to settle to the bottom of the tank due to gravity.

Secondary Treatment (Biological treatment)

- Purpose of biological treatment is to convert **organic wastes** into **stabilized compounds using microorganisms**.
- Analogous to **self purification in streams** but at an accelerated rate. Hence, our aim is to provide environment required for growth of microbes

Activated sludge process

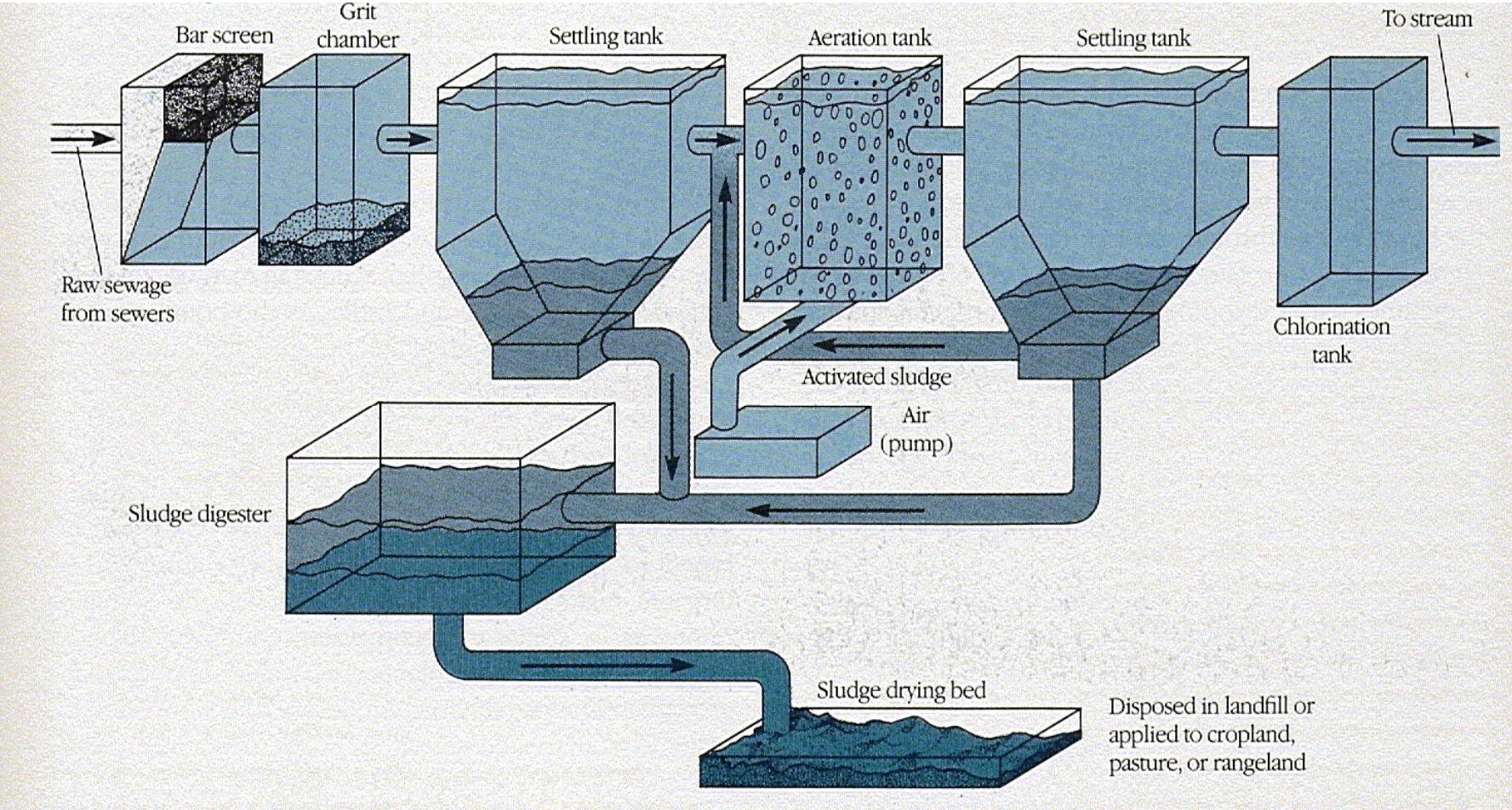
- **Microorganisms** are mixed thoroughly with the organic compounds in wastewater under conditions that **stimulate their growth** through **use of the organic compounds as substrate**.
- The microorganisms grow and are mixed by the **agitation of the air**, the individual organisms flocculate to form an active mass of microbes (biologic floc) called activated sludge.

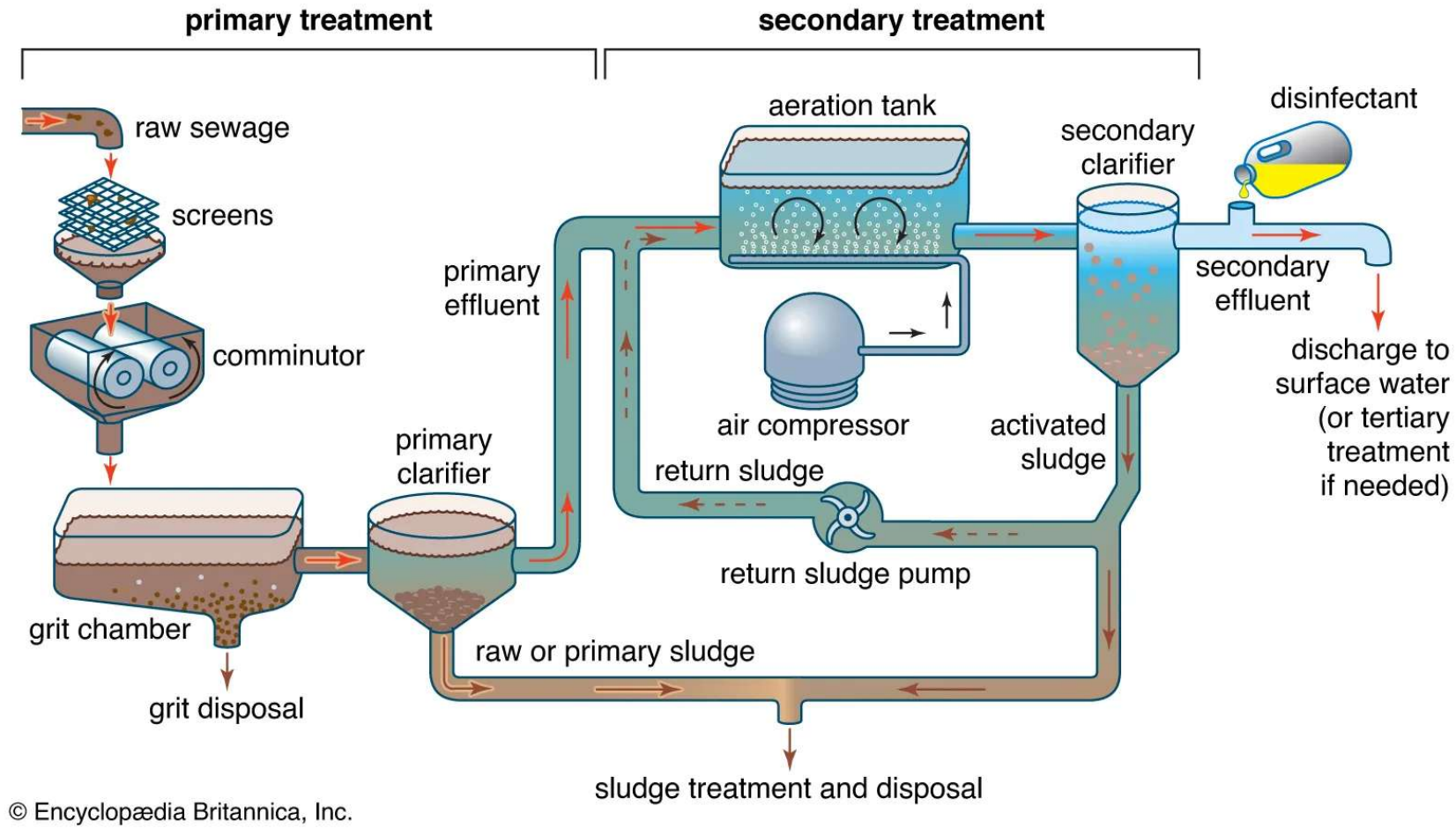
Activated sludge process





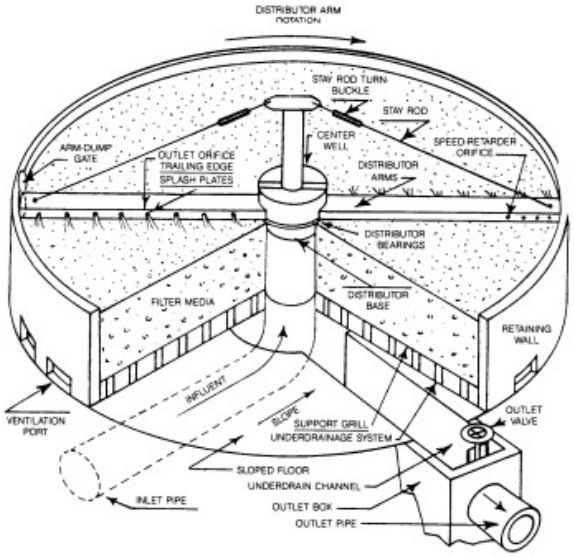
Conventional Sewage Treatment process




primary treatment
secondary treatment
raw sewage
screens
comminutor
grit chamber
grit disposal
primary clarifier
primary effluent
aeration tank
air compressor
return sludge
return sludge pump
activated sludge
secondary clarifier
disinfectant
secondary effluent
discharge to surface water (or tertiary treatment if needed)
raw or primary sludge
sludge treatment and disposal
© Encyclopædia Britannica, Inc.

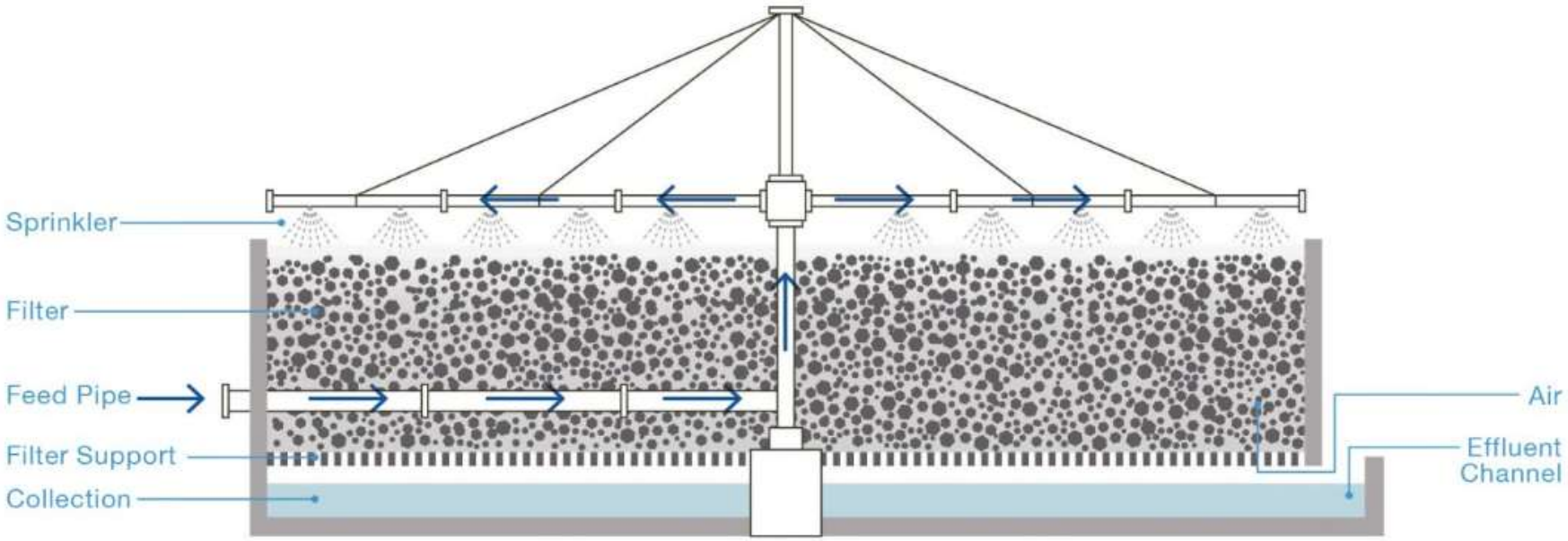
Trickling filters

- Trickling filters (TFs) are used to **remove organic matter from wastewater**. The TF is an aerobic treatment system that utilizes microorganisms attached to a medium to remove organic matter from wastewater.
- Tank with rock packing or plastic packing. Wastewater is sprayed on top of packing and trickles down getting biofilm treatment in the process.



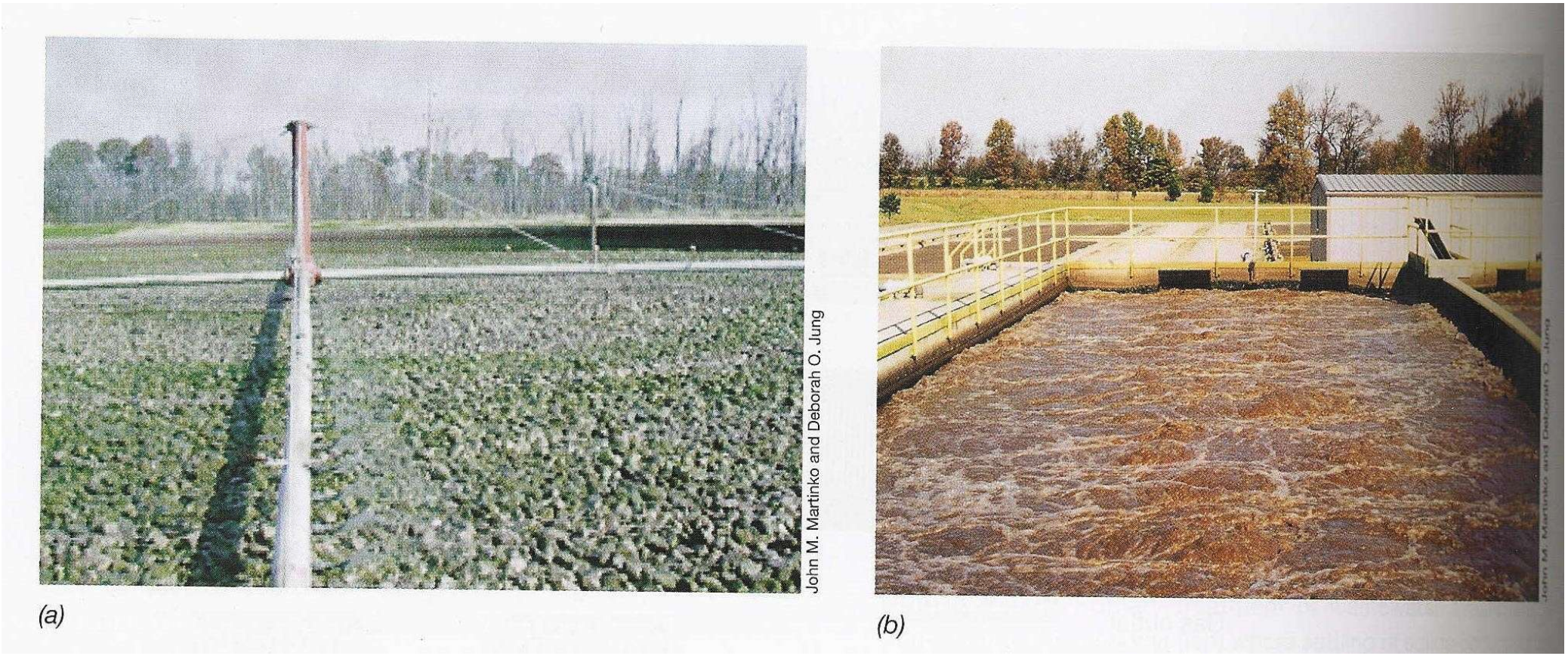
Trickling filter

- TFs enable **organic material in the wastewater to be adsorbed by a population of microorganisms** (aerobic, anaerobic, and facultative bacteria; fungi; algae; and protozoa) **attached to the medium** as a **biological film or slime layer** (approximately 0.1 to 0.2 mm thick).
- As the **wastewater flows over the medium**, **microorganisms already in the water gradually attach themselves to the rock, slag, or plastic surface and form a film**. The **organic material** is then degraded by the **aerobic microorganisms** in the outer part of the slime layer.
- As the **layer thickens through microbial growth**, **oxygen cannot penetrate the medium face, and anaerobic organisms develop**. As **the biological film continues to grow**, the **microorganisms near the surface lose their ability to cling to the medium, and a portion of the slime layer falls off the filter**. This process is known as **sloughing**. The **sloughed solids are picked up by the underdrain** system and **transported to a clarifier** for removal from the wastewater.



Sectional View of a Trickling Filter

Trickling Filter and Aeration Basin for Wastewater Treatment



Waste Water Treatment Plant, Indore



https://youtu.be/FvPakzqM3h8?si=THrM6SdTR-UUL13L

SAVE WATER

SAVE LIFE

